GuardianPixel: Enhancing Data Security through AI-based Steganography

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PRESENTATION OUTLINE

1. Introduction
2. Literature Survey
3. Implementation Plan
4. Application
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INTRODUCTION: SECURING DATA

Cryptography:

- **Protects Content:** Successfully scrambles the actual message.
- **Leaves a Trace:** Results in a visibly encrypted, unreadable file.
- **Draws Attention:** The scrambled nature makes the presence of a secret obvious.
- **Creates a Target:** Attackers know exactly what to intercept and attack.

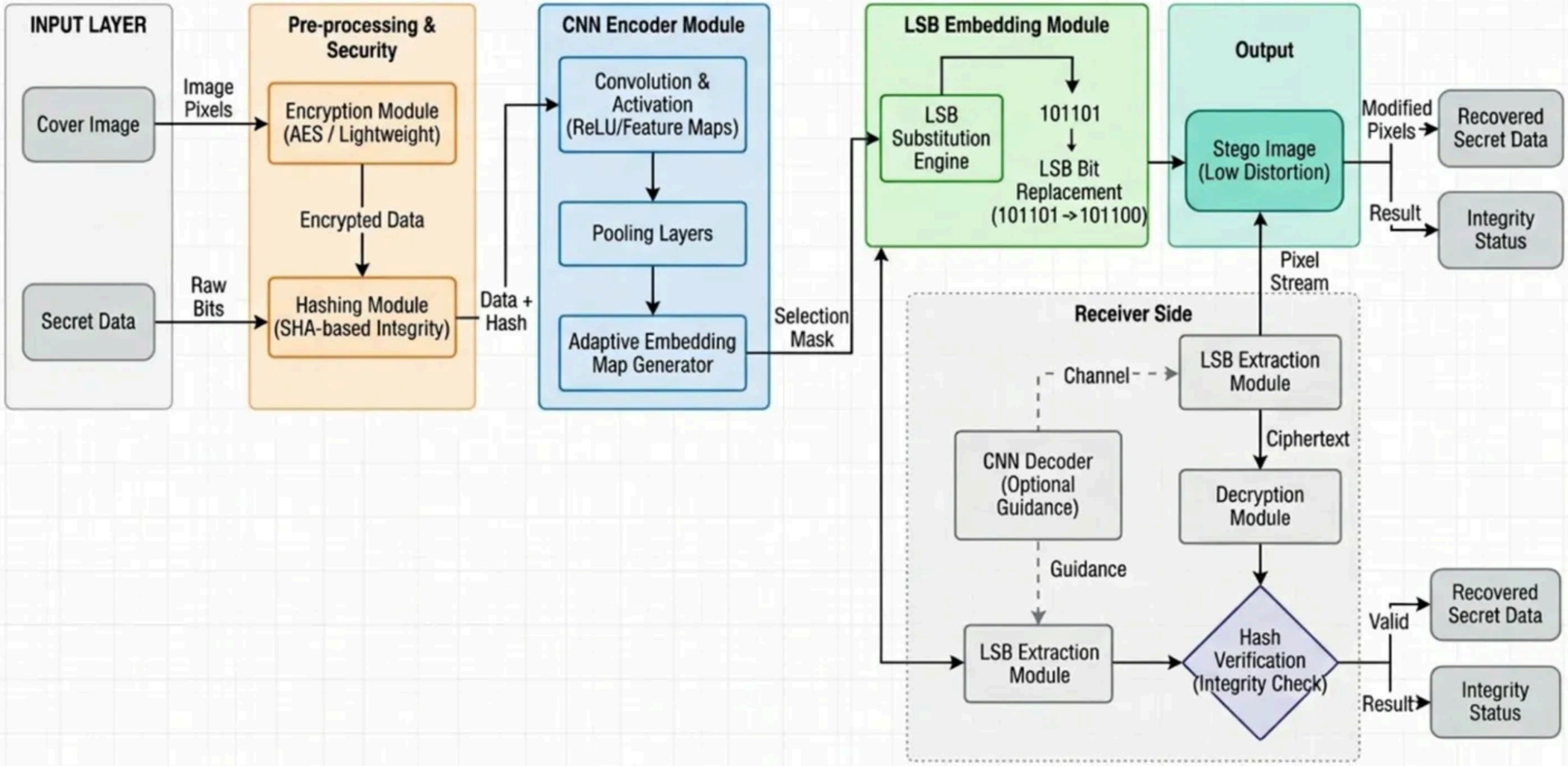
Steganography:

- **Protects Existence:** Completely conceals the secret data within ordinary digital media (images).
- **Leaves No Trace:** The file looks and behaves exactly like a normal image.
- **Deflects Attention:** Hides the communication entirely from unauthorized viewers.
- **Removes the Target:** Attackers cannot attack a secret they don't know exists.

INTRODUCTION: THE AI ADVANTAGE

- **The Traditional Flaw:** Standard steganography alters pixels blindly, often distorting regions and leaving statistical traces that scanners can catch.
- **The GuardianPixel Solution:** We introduce Machine Learning as an intelligent, content-aware capacity filter.
- **Smart Region Selection:** The AI model scans the cover image to identify ‘high-entropy’ zones—areas with complex textures, rough edges, and high variance.
- **Secure Embedding:** Encrypted data is embedded only in these complex regions, exploiting the limits of human visual perception to completely mask the changes.

EXISTING SYSTEM ARCHITECTURE



SUMMARY OF LITERATURE SURVEY

Reference	Technique	Key Features	Advantages	Limitations
Al-Rawashdeh et al. [1]	Edge Detection + CNN	Canny/Sobel edge detection, CNN embedding/extraction	8 BPP capacity, strong robustness, resistance to steganalysis	Dependence on conventional edge detectors, limited to 3 datasets
Xiao et al. [2]	SRIS-Net	LPFDD multi-domain, GLEM, DMSubNet, dual-task discriminator	PSNR 43.55 dB, SSIM 0.999, 96 BPP capacity, high undetectability	Complex architecture, high training requirements
Rafat and Sajjad [3]	Reversible LSB with Randomization	Triple image generation, SHA-256 key, XOR combination	PSNR > 51 dB, resistance to known-cover attacks, adherence to Kerckhoff	Triple storage requirement, grayscale only
Tafreshi et al. [4]	CNN-Based SSIS	Sinusoidal PN classes, CNN classification, Fourier domain	PSNR 44.5 dB, geometric robustness, enhanced security	CNN training required, JPEG compression vulnerability.
Judy and Khilar [5]	MOAF	Entropy-based adaptive compression/encryption, SHA-256 key generation	PSNR 81.31 dB, SSIM 1.0, adaptive selection	Limited algorithms, medical images only

PROBLEM STATEMENT

Traditional steganography is detectable, non-adaptive, and lacks strong security due to insufficient integration of hashing, encryption, and data integrity mechanisms.

SCOPE

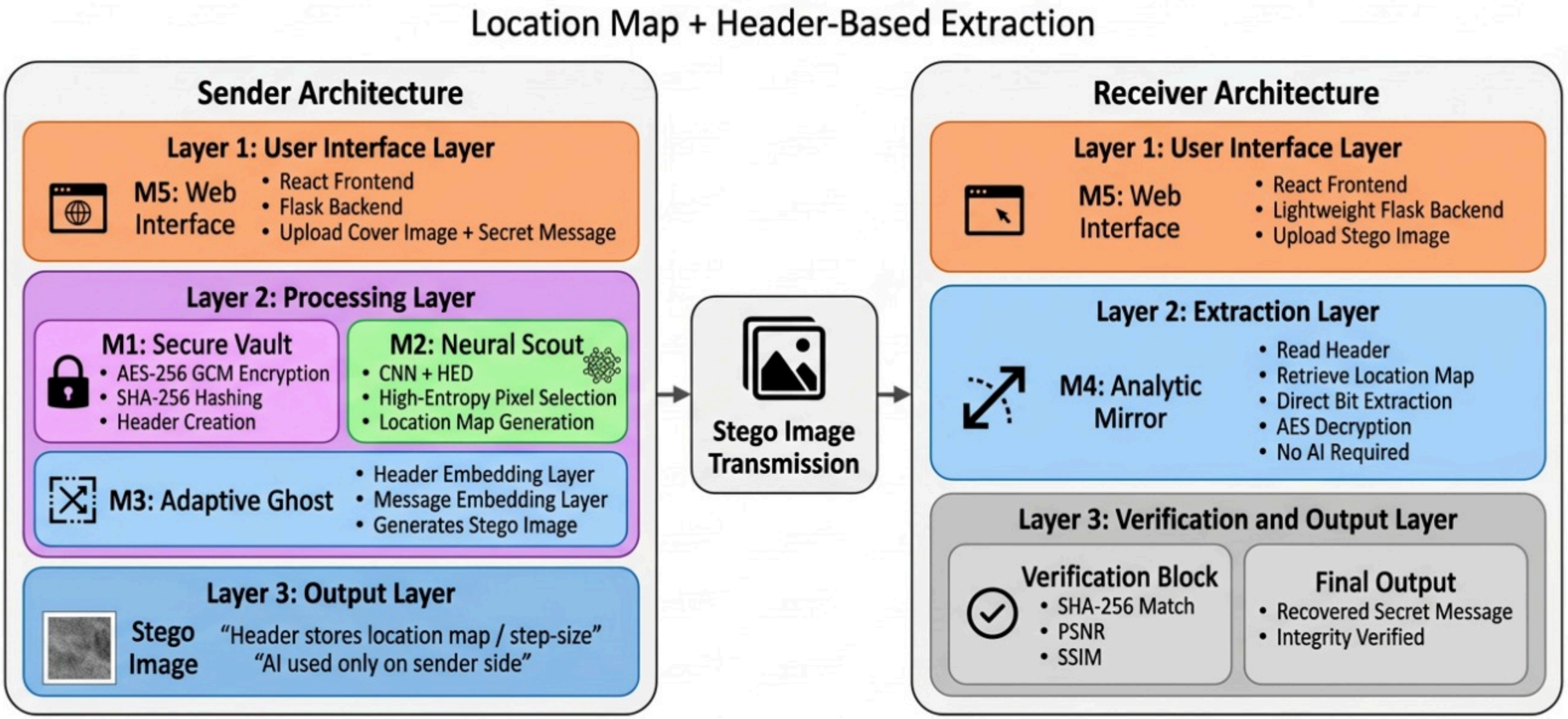
- Hybrid system: CNN + adaptive LSB embedding
- Integration of hashing + encryption for enhanced security
- Secure image-based steganography system

OBJECTIVES

- Apply AES-256 encryption + SHA-256 hashing
- Generate location map via CNN
- Perform adaptive embedding
- Ensure quality (PSNR, SSIM) & integrity

SYSTEM ARCHITECTURE

SOLUTION 1



Mathematical Model & Algorithms

Cryptography & Integrity (The Secure Vault)

AES-256 GCM Encryption → Converts message into secure ciphertext
Provides: Confidentiality Authentication

Data Integrity (SHA-256 Hashing) → Ensures data integrity
hash matches → data is authentic
If mismatch → data tampered

The "Instruction" Header (The Map) → Secure binary string embedded in a fixed “Safe Zone”

Mathematical Model & Algorithms

AI-Driven Selection & Embedding (The Neural Scout)

HED CNN (Holistically-Nested Edge Detection)	Converts message into secure ciphertext Provides: Confidentiality Authentication
Selection Logic	Only high-entropy (complex) regions are chosen, as changes in these areas are less noticeable
Double-Layer Embedding Approach:	Layer 1 (Map Storage) Layer 2 (Secret Data)
Efficient Extraction:	The receiver directly uses the stored map to locate and extract the hidden data without running the AI model again

HARDWARE AND SOFTWARE DETAILS

Core AI & Processing Engine	• Python 3.10+ for AI & image processing • PyTorch / TensorFlow for HED CNN • OpenCV for LSB embedding & pixel operations • NumPy for fast matrix computations
System Integration (Optional)	• Flask for connecting AI modules via API • React.js for frontend dashboard
Database & Storage	• PostgreSQL for storing maps & hashes • Local storage for cover & stego images
Hardware Requirements	• CPU: Intel i5/i7 or Ryzen 5/7 • RAM: 8GB (16GB recommended) • GPU: For faster CNN processing • Storage: SSD preferred

INPUT AND OUTPUT SPECIFICATION

INPUTS

- Cover Image: High-resolution lossless formats (PNG, TIFF) to avoid compression artifacts.
- Secret Data: Text strings or small files.
- User Keys: AES-256 Encryption key and Header Access key.

OUTPUTS

- Stego-Image: Visually indistinguishable from the original cover.
- Security Report: Displays PSNR (Peak Signal-to-Noise Ratio) and SSIM (Structural Similarity Index).
- Extracted Message: Verified via SHA-256 hash match.

Performance Evaluation Parameters

Imperceptibility : High image quality maintained with PSNR > 40 dB and SSIM > 0.95 (minimal visible distortion)

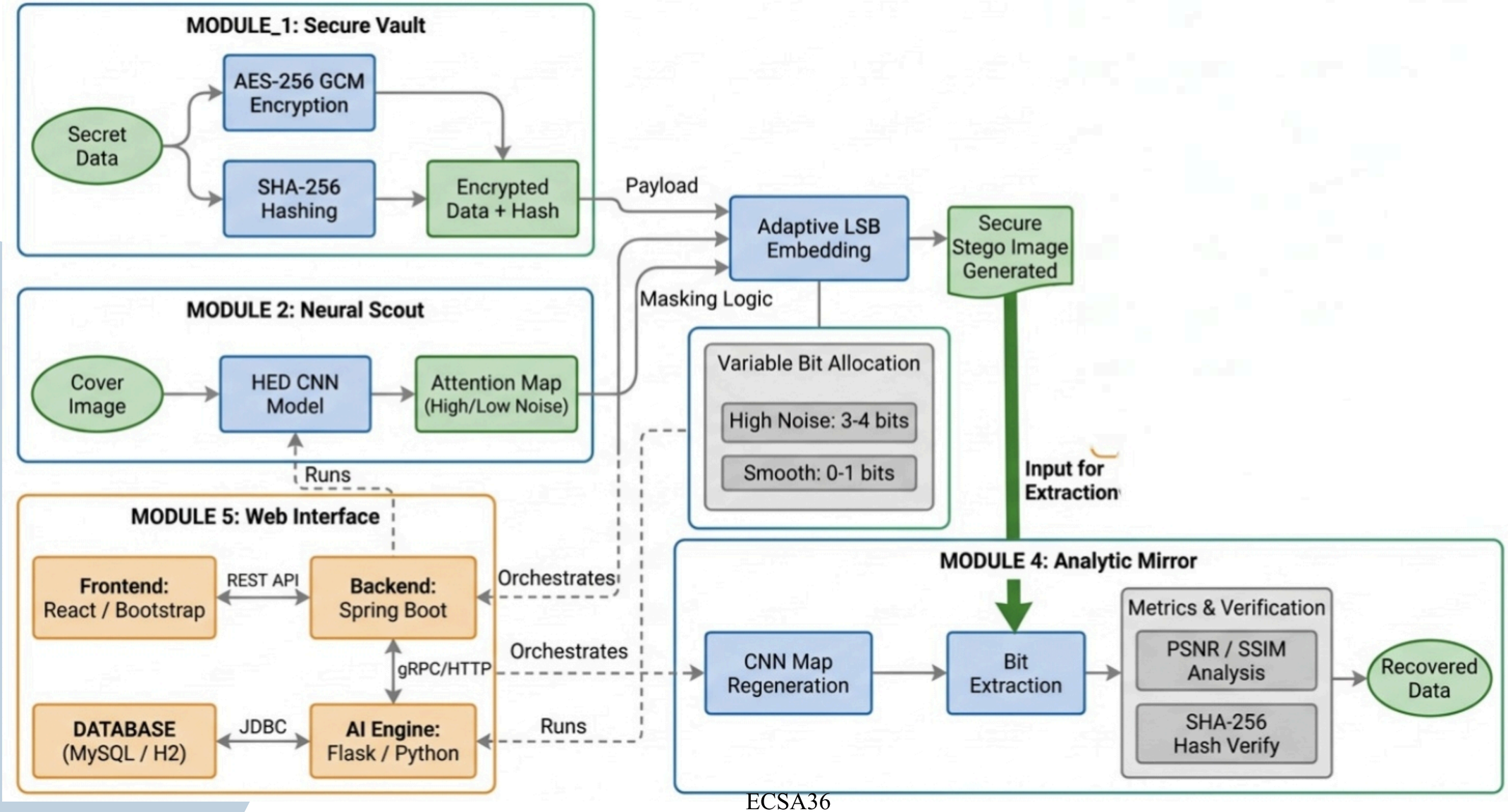
Robustness: Strong resistance against statistical attacks like Chi-square and histogram analysis

Computational Efficiency: Optimized performance with fast AI-based pixel mapping and quicker extraction process

Payload Capacity: Efficient data embedding with high bits-per-pixel (BPP) while preserving image quality

SYSTEM ARCHITECTURE

SOLUTION 2



Comparison: Approach 1 vs Approach 2

Feature	Approach 1: Location Mapping (Map)	Approach 2: Synchronous AI (Mirror)	Why Approach 1 Wins
Receiver Logic	• Reads small binary header • Direct pixel access	• Receiver runs full HED CNN • High computation required	• No GPU needed • Lightweight & efficient
System Stability	• Fixed coordinates from sender • Deterministic output	• Sensitive to small CNN variations • Risk of data loss	• No floating-point errors • Highly reliable
Extraction Speed	• Instant extraction using map • No AI delay	• Slow due to AI inference • Extra processing time	• Faster performance • Better user experience
Dependency	• AI only at sender side • Simple extraction logic	• Requires full AI setup on both sides • Heavy environment	• Works on mobile/web • High portability
Flexibility	• Select top N best pixels • Adjustable embedding	• Limited to real-time AI output • Less control	• Better control over security • Optimized capacity

Proposed Solution

Smarter at the source, simpler at the destination.

Offload all AI computation to the sender, making the receiver a zero-ML thin client capable of running on a device

Embed the AI-generated coordinate map directly inside the stego image, eliminating sync errors and ensuring consistent extraction across all devices.

Use HED deep learning to intelligently select pixels where the human eye is least sensitive, going far beyond traditional LSB steganography.

Design the full system around asymmetric processing heavy lifting on send, near-instant on receive making it viable for real-world, low-resource deployment.

APPLICATIONS

- Secure Military and Defense Communication
- Medical Data Security and Privacy
- Digital Watermarking and Copyright Protection
- Covert Communication Systems
- Cloud Storage and Data Protection
- Secure Financial Transactions
- Law Enforcement and Digital Forensics
- Satellite Image Data Security

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